THE MAP FROM A LINEARLY TOPOLOGIZED RING TO ITS COMPLETION IS TAUT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof refers to a comparison with an earlier lemma for the identification of the kernels; that reference is carried as a navigation stub.

Editorial note. The counted spans carry 1 navigation stub; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- The completion of a linearly topologized ring as the limit of the quotients by a fundamental system of open ideals, with the limit topology. Standard, recalled at proof lines 1346-1350. Supplies the fundamental system of open ideals of the completion used in the proof.
- The kernel of the map to a quotient is the closure of the extension of the corresponding ideal. Same chapter, Lemma lemma-closure-image-ideal, invoked at proof lines 1351-1353. Identifies the kernels of the structure maps, which is the condition of tautness.
- Tautness of a continuous map of linearly topologized rings. Same chapter, the definition of a taut map, in force in the statement. Fixes what has to be shown.

Source reference. Original lemma 0AMZ: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1165–1165. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let A be a linearly topologized ring. The map $A \to A^\wedge$ from A to its completion is taut.*

Proof. Let I_λ be a fundamental system of open ideals of A. Recall that $A^\wedge = \lim A/I_\lambda$ with the limit topology, which means that the kernels $J_\lambda = \mathrm{Ker}(A^\wedge \to A/I_\lambda)$ form a fundamental system of open ideals of $A^\wedge$. Since J_λ is the closure of $I_\lambda A^\wedge$ (compare with Lemma 0AMZ) we conclude. □

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